

Time-independent Schrodinger equation

Consider the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = H \Psi(x, t)$$

Let it be the case that

$$\Psi(x, t) = \sum_k e^{-iE_k t/\hbar} \phi_k(x)$$

Then for all k

$$i\hbar \frac{d}{dt} e^{-iE_k t/\hbar} \phi_k(x) = E_k e^{-iE_k t/\hbar} \phi_k(x)$$

and by the Schrodinger equation we have

$$E_k e^{-iE_k t/\hbar} \phi_k(x) = H [e^{-iE_k t/\hbar} \phi_k(x)]$$

If H is independent of t then we can rearrange as

$$E_k e^{-iE_k t/\hbar} \phi_k(x) = e^{-iE_k t/\hbar} H \phi_k(x)$$

The $e^{-iE_k t/\hbar}$ cancel leaving the time-independent Schrodinger equation

$$E_k \phi_k(x) = H \phi_k(x)$$

By linearity of H we also have

$$\sum_k E_k \phi_k(x) = H \sum_k \phi_k(x)$$

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